Forces and motion

Mark scheme and worked solutions for 45 questions: 15 multiple choice, 30 structured.

1 · Part 1

Paper 1 & 2 · Units · [1]

Answer: A

Explanation

Gravitational field strength is the force on each kilogram of mass, so its unit is the newton per kilogram, N/kg. N is the unit of force (weight), kg/N is the inverse of the correct unit, and m/s is the unit of speed or velocity.

2 · Part 1

Paper 2 only · Units · [1]

Answer: C

Explanation

The newton metre (Nm) is the unit of the moment (turning effect) of a force. Force is measured in N, momentum in kg m/s and pressure in Pa (N/m^2), so A, B and D are wrong.

3 · Part 1

Paper 1 & 2 · Movement and position · [1]

Answer: D

Worked solution

average speed = distance moved / time taken = 4500 m / 900 s = 5.0 m/s. Always change km to m and minutes to s before using the equation when the answer must be in m/s.

4 · Part 1

Paper 1 & 2 · Movement and position · [1]

Answer: B

Worked solution

Change in velocity = 24 − 6.0 = 18 m/s. Rearranging a = (v − u) / t gives t = (v − u) / a = 18 / 3.0 = 6.0 s. Check: 6.0 + (3.0 × 6.0) = 24 m/s.

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5 · Part 1 Paper 1 & 2 Movement and position [1]

Answer: A

Worked solution

The graph is a sloping straight line from the origin, so the area under it is a triangle with base 5.0 s and height 6.0 m/s. Area = ½ × base × height = ½ × 5.0 × 6.0 = 15 m. Check: the average velocity is (0 + 6.0) / 2 = 3.0 m/s, and 3.0 m/s × 5.0 s = 15 m.

6 · Part 1 Paper 1 & 2 Units [6]

Mark scheme

(a) **[2]**

- mass: kg; force: N; velocity: m/s; time: s — all four correct (2)
- two or three correct (1)

ALLOW kilogram, newton, metre/second (or metre per second), second written in words

(b) **[2]**

- 18 km = 18 000 m AND 1 h = 3600 s (1)
- 18 000 / 3600 = 5 (m/s) (1)

ALLOW 18 × 1000 / 3600 = 5 OR 18 / 3.6 = 5

(c) **[2]**

- acceleration = change in velocity / time taken (1)
- (m/s) ÷ s gives m/s^2 (1)

ALLOW "metres per second per second" for the unit

Worked solution

(a) Mass is measured in kilograms (kg), force in newtons (N), velocity in metres per second (m/s) and time in seconds (s).

(b) 18 km is 18 × 1000 = 18 000 m. One hour is 60 × 60 = 3600 s. So 18 km/h = 18 000 m / 3600 s = 5 m/s.

(c) Acceleration is the change in velocity divided by the time taken. Velocity is in m/s and time is in s, so the unit is (m/s) / s = m/s^2. This means the velocity changes by a number of metres per second every second.

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7 · Part 1 Paper 2 only Units [4]

Mark scheme

(a) **[1]**

- 360 g = 0.36 kg, so 360 g m/s = 0.36 kg m/s (1)

(b) **[1]**

- 0.36 kg m/s (1)

(c) **[2]**

- Dina's value converted correctly: 0.45 kg m/s (1)

ALLOW Amir's value converted to 3.6×10^4 g cm/s instead

- Dina's trolley has more momentum, because 0.45 kg m/s is greater than 0.36 kg m/s (1)

ECF the conclusion from a wrongly converted value

Worked solution

(a) There are 1000 g in 1 kg, so 360 g = 0.36 kg. The velocity unit (m/s) is the same, so 360 g m/s = 0.36 kg m/s, the same as Amir's value.

(b) There are 100 cm in 1 m, so 36 kg cm/s = 36 / 100 kg m/s = 0.36 kg m/s.

(c) Convert both parts of Dina's unit. 4.5×10^4 g = 45 kg, so 4.5×10^4 g cm/s = 45 kg cm/s. Then 45 kg cm/s = 45 / 100 = 0.45 kg m/s. Since 0.45 kg m/s is greater than 0.36 kg m/s, Dina's trolley has more momentum. Values can only be compared fairly when they are in the same unit.

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8 · Part 1 Paper 1 & 2 Movement and position [9]

Mark scheme

(a) **[1]**

- average speed = distance moved / time taken (1)

ALLOW any correct rearrangement or standard symbols

(b) **[3]**

- 6.0 km = 6000 m AND 10 minutes = 600 s (1)
- average speed = 6000 / 600 (1)
- 10 m/s (1)

(c) **[3]**

- total distance = 15 km = 15 000 m AND total time = 30 minutes = 1800 s (1)
- average speed = 15 000 / 1800 (1)
- 8.3 m/s (1)

ALLOW 8.33 m/s

REJECT a total time of 25 minutes (stop left out) for the first mark; ECF the final answer from it

(d) **[2]**

- the bus is stationary during stage 2 OR the bus travels no distance during the 5.0 minute stop (1)
- so the total time increases but the total distance does not, so distance / time is smaller (1)

ALLOW the same distance travelled in a longer time

ALLOW a comparison using values: stage 3 average speed is 10 m/s, which is more than 8.3 m/s

Worked solution

(a) average speed = distance moved / time taken.

(b) Change units first: 6.0 km = 6000 m and 10 minutes = 10 × 60 = 600 s. Average speed = 6000 / 600 = 10 m/s.

(c) The whole journey includes the stop. Total distance = 6.0 + 9.0 = 15 km = 15 000 m. Total time = 10 + 5.0 + 15 = 30 minutes = 1800 s. Average speed = 15 000 / 1800 = 8.3 m/s. Check: 8.33 m/s × 1800 s = 15 000 m.

(d) In stage 3 the average speed is 9000 m / 900 s = 10 m/s. For the whole journey the bus spends 5 minutes standing still. This adds time but no distance, so the average speed for the whole journey is lower.

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9 · Part 1

Paper 1 & 2 | Movement and position | [9]

Mark scheme

(a) **[2]**

- independent variable: height of the (raised end of the) ramp (1)
- dependent variable: average speed of the car OR time taken for the car to roll down the ramp (1)

(b) **[1]**

- any one of: same toy car OR same mass of car; same distance travelled down the ramp; same ramp OR same ramp surface; same starting position; car released from rest each time (1)

REJECT "same height"

(c) **[6]**

Indicative content; award one mark for each point, max 6:

- raise one end of the ramp on the books and measure the height of the raised end (above the floor) with the metre rule (1)
- mark a start line and a finish line on the ramp and measure the distance between them with the metre rule (1)

ALLOW use the full 1.2 m length of the ramp

- release the car from rest at the start line, without pushing it (1)
- start the stopwatch when the car is released and stop it when the car reaches the finish line (1)
- calculate average speed = distance / time (1)
- repeat each height at least three times and calculate a mean time (1)
- change the height by adding or removing books and repeat for at least five different heights (1)
- identify and ignore anomalous results OR repeat any anomalous results (1)

IGNORE control variables already credited in (b)

Worked solution

(a) The student changes the height of the ramp, so that is the independent variable. The student measures the time and uses it to find the average speed, so average speed (or time) is the dependent variable.

(b) Keep everything else the same so that only the height affects the speed, for example use the same toy car each time.

(c) A good method: put the ramp on a few books and measure the height of the raised end with the metre rule. Mark a start line and a finish line on the ramp and measure the distance between them, for example 1.0 m. Hold the car at the start line and let it go without pushing. Start the stopwatch as you let go and stop it as the car passes the finish line. Repeat three times at this height and find the mean time, ignoring any anomalous times. Work out average speed = distance / mean time. Add or remove books to get at least five different heights and repeat the whole process. Repeats and a mean make the results more reliable.

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10 · Part 1 Paper 1 & 2 Movement and position [8]

Mark scheme

(a) **[1]**

- 3.9 s OR trial 3 (1)

(b) **[2]**

- (2.4 + 2.6 + 2.5 + 2.5) / 4 OR 10.0 / 4 (1)
- 2.5 s (1)

REJECT 2.78 s (anomaly included) for both marks

(c) **[2]**

- average speed = 5.0 / 2.5 (1)
- 2.0 m/s (1)

ECF from (b)

(d) **[1]**

- any one of: stopwatch stopped too late OR started too early (reaction time); ball rolled more slowly because it was rolled more gently; ball hit something or rolled off a straight line (1)

(e) **[2]**

- the error in starting and stopping the stopwatch (reaction time) is about the same size each time (1)
- this error is a smaller fraction (percentage) of the longer time measured over 5.0 m (1)

ALLOW the time over 0.50 m is too short to measure accurately with a stopwatch, as the reaction time is a large part of it, for the second mark

Worked solution

(a) Trial 3 (3.9 s) is much longer than the other four times, which are all between 2.4 s and 2.6 s, so it is anomalous.

(b) Leave out 3.9 s. Mean = (2.4 + 2.6 + 2.5 + 2.5) / 4 = 10.0 / 4 = 2.5 s.

(c) average speed = distance moved / time taken = 5.0 m / 2.5 s = 2.0 m/s. Check: 2.0 m/s × 2.5 s = 5.0 m.

(d) The friend may have stopped the stopwatch late, or the ball may have been rolled more gently in that trial.

(e) A person takes a fraction of a second to react when starting and stopping a stopwatch. This error is about the same each time. Over 5.0 m the time is about 2.5 s, so the error is a small part of it. Over 0.50 m the time would be only about 0.25 s, and the reaction-time error would be a large part of the measurement, so the speed would be much less accurate.

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11 · Part 1 Paper 1 & 2 Movement and position [8]

Mark scheme

(a) **[1]**

- acceleration = change in velocity / time taken (1)

ALLOW a = (v − u) / t

(b) **[2]**

- acceleration = (8.4 − 0) / 2.0 (1)
- 4.2 m/s^2 (1)

(c) **[2]**

- the first sprinter (has the greater acceleration) (1)
- because she has the same change in velocity in less time OR the second sprinter's acceleration is 8.4 / 2.4 = 3.5 m/s^2, which is less than 4.2 m/s^2 (1)

ECF from (b) for a numerical comparison

(d) **[3]**

- −1.5 = (v − 10.5) / 3.0 OR change in velocity = −1.5 × 3.0 = −4.5 m/s (1)
- v = 10.5 − 4.5 (1)
- 6.0 m/s (1)

ALLOW a decrease in velocity of 1.5 m/s every second, so 10.5 − (3 × 1.5), for the first two marks

Worked solution

(a) acceleration = change in velocity / time taken.

(b) She starts from rest, so the change in velocity is 8.4 − 0 = 8.4 m/s. Acceleration = 8.4 / 2.0 = 4.2 m/s^2.

(c) Both sprinters gain the same velocity, 8.4 m/s. The first does it in 2.0 s and the second takes longer, 2.4 s. Dividing the same change in velocity by a longer time gives a smaller acceleration: 8.4 / 2.4 = 3.5 m/s^2, compared with 4.2 m/s^2. So the first sprinter has the greater acceleration.

(d) An acceleration of −1.5 m/s^2 means her velocity drops by 1.5 m/s every second. In 3.0 s it drops by 1.5 × 3.0 = 4.5 m/s. Her velocity is 10.5 − 4.5 = 6.0 m/s. Check: (6.0 − 10.5) / 3.0 = −1.5 m/s^2.

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12 · Part 1 Paper 1 & 2 Movement and position [8]

Mark scheme

(a) **[1]**

- acceleration (1)

(b) **[3]**

- acceleration = change in velocity / time taken OR gradient = change in velocity / change in time (1)
- = 18.0 / 30 (1)
- 0.60 m/s^2 (1)

ALLOW any pair of points from 0 s to 30 s, e.g. 6.0 / 10

(c) **[2]**

- 0 (m/s^2) (1)
- the velocity stays constant at 18.0 m/s OR the gradient of the graph is zero (horizontal line) (1)

(d) **[2]**

- the second train has a greater acceleration (1)
- its velocity increases by more each second OR it reaches a higher velocity in the same time (1)

Worked solution

(a) The gradient of a velocity–time graph is the change in velocity divided by the change in time, which is the acceleration.

(b) Between 0 s and 30 s the velocity rises from 0 to 18.0 m/s. Gradient = 18.0 / 30 = 0.60 m/s^2. The line is straight, so any two points give the same answer: 6.0 / 10 = 0.60 m/s^2.

(c) From 30 s to 50 s the velocity stays at 18.0 m/s. The change in velocity is zero, the line is horizontal and the gradient is zero, so the acceleration is 0.

(d) A steeper gradient means a larger acceleration. The second train's velocity increases faster, so after any time it is moving faster than the first train.

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13 · Part 1 Paper 1 & 2 Movement and position [9]

Mark scheme

(a) **[1]**

- the distance travelled (1)

(b) **[4]**

- 0 s to 10 s: ½ × 10 × 8.0 = 40 m (1)
- 10 s to 40 s: 30 × 8.0 = 240 m (1)
- 40 s to 50 s: ½ × 10 × 8.0 = 40 m (1)
- total = 320 m (1)

ALLOW trapezium method: ½ × (50 + 30) × 8.0 = 320 m for full marks

(c) **[2]**

- average speed = 320 / 50 (1)
- 6.4 m/s (1)

ECF from (b)

(d) **[2]**

- the scooter travels slower than 8.0 m/s while it is speeding up (0 s to 10 s) and slowing down (40 s to 50 s) (1)
- so it travels less distance in 50 s than it would at a constant 8.0 m/s OR its speed is only 8.0 m/s for part of the time (1)

ALLOW at 8.0 m/s for 50 s it would travel 400 m, which is more than 320 m

Worked solution

(a) The area under a velocity–time graph gives the distance travelled.

(b) Split the area into three shapes. Speeding up (0–10 s) is a triangle: ½ × 10 × 8.0 = 40 m. Constant speed (10–40 s) is a rectangle: 30 × 8.0 = 240 m. Slowing down (40–50 s) is a triangle: ½ × 10 × 8.0 = 40 m. Total = 40 + 240 + 40 = 320 m.

(c) average speed = distance moved / time taken = 320 m / 50 s = 6.4 m/s. Check: 6.4 × 50 = 320 m.

(d) The scooter only moves at 8.0 m/s for 30 s. While speeding up and slowing down it moves more slowly, so the average over the whole 50 s is less than 8.0 m/s.

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14 · Part 1 Paper 1 & 2 Movement and position [9]

Mark scheme

(a) **[3]**

- $0^2 = 15^2 + (2 \times a \times 18)$ (1)
- $a = -225 / 36$ (1)
- -6.25 m/s^2 (1)

ALLOW -6.3 m/s^2

ALLOW 6.25 m/s^2 or 6.3 m/s^2 without the minus sign if the answer is described as a deceleration

(b) **[2]**

- $t = (0 - 15) / (-6.25)$ OR $t = 15 / 6.25$ (1)
- 2.4 s (1)

ECF from (a)

(c) **[2]**

- $0^2 = 30^2 + (2 \times (-6.25) \times s)$ OR $s = 900 / 12.5$ (1)
- $s = 72$ (m) (1)

ALLOW $a = 6.3$ m/s^2 from (a) giving $s = 900 / 12.6 = 71.4$ m, stated as approximately 72 m, for both marks

ALLOW ECF of any other value of a from (a) only for the first mark

(d) **[2]**

- when the car stops (final speed zero) the distance is proportional to (initial speed)^2, $s = u^2 / (2 \times$ deceleration) (1)
- doubling the initial speed makes (initial speed)^2 four times larger, so the distance is four times larger (when the acceleration is the same) (1)

ALLOW a numerical explanation: $15^2 = 225$ and $30^2 = 900$, which is 4 times larger, so 18 m becomes 72 m

Worked solution

(a) The car stops, so the final speed $v = 0$. Substitute: $0 = 15^2 + (2 \times a \times 18)$, so $0 = 225 + 36a$ and $a = -225 / 36 = -6.25$ m/s^2. The minus sign shows the car is slowing down.

(b) acceleration = change in velocity / time taken, so $t = (0 - 15) / (-6.25) = 2.4$ s. Check: the average speed while braking is $(15 + 0) / 2 = 7.5$ m/s, and $7.5 \times 2.4 = 18$ m, which matches.

(c) $0 = 30^2 + (2 \times (-6.25) \times s)$, so $0 = 900 - 12.5s$ and $s = 900 / 12.5 = 72$ m.

(d) With $v = 0$ the equation becomes $s = u^2 / (2 \times 6.25)$, so the braking distance depends on the square of the initial speed. Doubling u multiplies u^2 by $2 \times 2 = 4$, so 18 m becomes $4 \times 18 = 72$ m.

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15 · Part 1

Paper 1 & 2 Movement and position [9]

Mark scheme

(a) **[2]**

- acceleration = (60 − 0) / 24 (1)
- = 2.5 (m/s^2) (1)

(b) **[3]**

- $60^2 = 0^2 + (2 \times 2.5 \times s)$ (1)
- s = 3600 / 5.0 (1)
- 720 m (1)

(c) **[2]**

- average speed = 720 / 24 (1)
- 30 m/s (1)

ALLOW (0 + 60) / 2 = 30 m/s for both marks

ECF from (b)

(d) **[2]**

- the aircraft travels a longer distance (along the runway) before take-off (1)
- because with the same final speed, $s = v^2 / (2 \times a)$, so a smaller acceleration gives a larger distance OR it takes longer to reach 60 m/s, so it travels further (1)

Worked solution

(a) acceleration = change in velocity / time taken = (60 − 0) / 24 = 2.5 m/s^2.

(b) The aircraft starts from rest, so u = 0. Then $60^2 = 0 + (2 \times 2.5 \times s)$, so 3600 = 5.0s and s = 3600 / 5.0 = 720 m.

(c) average speed = distance moved / time taken = 720 / 24 = 30 m/s. Check: for uniform acceleration from 0 to 60 m/s the average speed is halfway, (0 + 60) / 2 = 30 m/s, which agrees.

(d) The aircraft still needs to reach 60 m/s. With a smaller acceleration it takes longer to reach this speed, and it is moving along the runway the whole time, so it covers a longer distance. The equation shows the same thing: $s = v^2 / (2 \times a)$, so a smaller a gives a larger s.

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1 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [1]

Answer: B

Worked solution

Ask of each quantity: does it make sense to give a direction? A force of 5 N to the left, a velocity of 3 m/s north and an acceleration of 2 m/s^2 downwards all need a direction, so they are vectors. Distance, mass, speed, time and energy have no direction, so they are scalars. Weight and displacement are vectors, but C and D each also contain a scalar.

2 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [1]

Answer: D

Explanation

The Earth's gravitational force pulls the satellite towards the centre of the Earth and keeps it in orbit. Electrostatic forces act between charged objects and magnetic forces between magnets or magnetic materials; neither keeps a satellite in orbit. There is no frictional force holding it in orbit; in space there is almost no air to cause friction.

3 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [1]

Answer: A

Worked solution

For forces along one line, choose one direction as positive. Taking right as positive: +40 − 25 − 10 = +5 N, so the resultant is 5 N to the right.

4 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [1]

Answer: C

Worked solution

In force = mass × acceleration, the force is the resultant (unbalanced) force. Friction opposes the motion, so subtract it from the pull first: 12 N − 2.0 N = 10 N. Then a = 10 N / 2.5 kg = 4.0 m/s^2. Check: 2.5 kg × 4.0 m/s^2 = 10 N.

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5 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [1]

Answer: A

Worked solution

Mass is the amount of matter, so it is the same everywhere: 3.0 kg. Weight = mass × gravitational field strength, so it changes with g: 3.0 kg × 4.0 N/kg = 12 N. Check: g falls from 10 to 4.0 N/kg (× 0.4), so the weight falls from 30 N to 12 N (× 0.4).

6 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [1]

Answer: D

Explanation

Tiredness increases the driver's reaction time, so the car travels further before the brakes are applied: the thinking distance increases, but once braking starts nothing about the car or road has changed. An icy road (A) and worn brakes (C) reduce the braking force or grip, and a heavy load (B) means a smaller deceleration for the same braking force; all three increase the braking distance and do not affect the thinking distance.

7 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [1]

Answer: B

Explanation

Elastic behaviour is the ability of a material to recover its original shape after the forces causing deformation have been removed, so returning to its original shape and length shows elastic behaviour. A describes Hooke's law, which is a different idea; a material can be elastic without obeying Hooke's law. C shows the band has not fully recovered its shape. D happens while it is stretched and says nothing about recovery.

8 · Part 2

Paper 2 only | Forces, movement, shape and momentum | [1]

Answer: C

Worked solution

momentum = mass × velocity, with mass in kg and velocity in m/s, gives kg m/s. 150 g ÷ 1000 = 0.150 kg, and 0.150 kg × 20 m/s = 3.0 kg m/s.

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9 · Part 2

Paper 2 only | Forces, movement, shape and momentum | [1]

Answer: A

Worked solution

Write the force as "object 1 pulls object 2". The partner is always "object 2 pulls object 1", of the same type, equal in size and opposite in direction. Earth pulls book (gravitational) pairs with book pulls Earth (gravitational).

10 · Part 2

Paper 2 only | Forces, movement, shape and momentum | [1]

Answer: D

Worked solution

Moment = force × perpendicular distance from the pivot. Using the principle of moments: F × 0.20 m = 6.0 N × 0.30 m, so F = 1.8 Nm / 0.20 m = 9.0 N. Check: the force is on the side closer to the pivot, so it must be larger than 6.0 N.

11 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [5]

Mark scheme

(a) **[3]**

- changes the speed of the ball (1)
- changes the direction of the ball (1)
- changes the shape of the ball (it is squashed while in contact with the racket) (1)

ALLOW changes its velocity for the first or second mark, but not both

(b) **[1]**

- air resistance (1)

ALLOW drag OR friction (with the air)

(c) **[1]**

- gravitational (force) (1)

ALLOW weight OR gravity

Worked solution

A force can change an object's speed, its direction of motion or its shape. The racket does all three: the ball slows down, stops, then speeds up in the opposite direction, and while it is in contact with the strings it is squashed out of shape. In the air, air resistance (a kind of friction) acts opposite to the ball's motion and the gravitational force of the Earth (its weight) pulls it downwards.

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12 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [6]

Mark scheme

(a) **[1]**

- a vector has size (magnitude) and direction, but a scalar has size (magnitude) only (1)

(b) **[2]**

- 130 N (1)
- to the left (1)

(c) **[1]**

- 1450 N (1)

(d) **[2]**

- distance is a scalar: it is the total length of the path travelled, 400 m (1)
- displacement is a vector: the straight-line distance from the start in a given direction, and she ends where she started, so it is zero (1)

Worked solution

Force is a vector, so direction matters. Along one line, subtract opposing forces: 1450 − 1320 = 130 N, in the direction of the larger force (left). When the forces are balanced, Team 2's pull must equal Team 1's, 1450 N. Distance counts every metre run, 400 m. Displacement only compares the finish position with the start position; they are the same point, so the displacement is zero.

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13 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [7]

Mark scheme

(a) **[2]**

- a = (v − u) / t = (15 − 0) / 6.0 (1)
- 2.5 m/s^2 (1)

(b) **[2]**

- F = m × a = 1200 × 2.5 (1)
- 3000 N (1)

ECF from (a)

(c) **[1]**

- 3600 N (1)

ECF answer to (b) + 600 N

(d) **[2]**

- air resistance (resistive force) increases as the speed increases (1)
- so the driving force must increase to keep the same resultant force (needed for the same acceleration, F = m × a) (1)

Worked solution

Acceleration is the change in velocity divided by the time: 15 m/s ÷ 6.0 s = 2.5 m/s^2. The resultant force is mass × acceleration: 1200 kg × 2.5 m/s^2 = 3000 N. The resultant force is the driving force minus the resistive force, so the driving force = 3000 + 600 = 3600 N. As the car speeds up, air resistance gets bigger. If the driving force stayed the same, the resultant force and the acceleration would fall, so the engine must push harder to keep the acceleration at 2.5 m/s^2.

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14 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [7]

Mark scheme

(a) **[2]**

- mass is the amount of matter in the spacecraft, which does not change with location (1)
- weight is the gravitational force on it, which depends on the gravitational field strength, which is smaller on the Moon (1)

(b) **[2]**

- $W = m \times g = 2000 \times 1.6$ (1)
- 3200 N (1)

(c) **[3]**

- resultant force = 4000 − 3200 = 800 N (1)
- $a = F / m = 800 / 2000 = 0.40\ m/s^2$ (1)
- upwards (so the spacecraft is slowing down) (1)

ECF from (b)

REJECT using 4000 N as the resultant force for the second mark

Worked solution

The spacecraft contains the same matter wherever it is, so its mass stays 2000 kg. Its weight is mass × gravitational field strength: 2000 kg × 1.6 N/kg = 3200 N, much less than on Earth. The upward engine force (4000 N) is bigger than the weight (3200 N), so the resultant force is 800 N upwards. Acceleration = 800 N ÷ 2000 kg = 0.40 m/s^2 upwards. Because the spacecraft is moving down but accelerating up, it is slowing down, which is what is needed for a soft landing.

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15 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [8]

Mark scheme

(a) **[1]**

- 44 m (1)

(b) **[3]**

- both distances increase as the speed increases (1)
- thinking distance is proportional to speed, e.g. it doubles from 6 m to 12 m when the speed doubles from 10 m/s to 20 m/s (1)
- braking distance increases more than in proportion to speed, e.g. it is four times bigger (8 m to 32 m) when the speed doubles OR nine times bigger (8 m to 72 m) when the speed triples (1)

(c) **[2]**

- the mass is larger, so for the same braking force the deceleration is smaller (a = F / m) (1)
- so the car travels further before it stops (1)

(d) **[2]**

- time = distance / speed = 6 / 10 (OR 12 / 20 OR 18 / 30) (1)
- 0.6 s (1)

ALLOW 0.60 s

Worked solution

Stopping distance = thinking distance + braking distance = 12 + 32 = 44 m. Thinking distance goes up in the same ratio as the speed (6, 12, 18 m), because the car travels at its full speed during the driver's reaction time. Braking distance rises much faster (8, 32, 72 m): doubling the speed makes it four times larger. A heavier car has more mass, so the same braking force gives a smaller deceleration and the car needs a longer distance to stop. During the thinking time the car moves at constant speed, so reaction time = thinking distance ÷ speed = 6 m ÷ 10 m/s = 0.6 s; the other rows give the same value.

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16 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [7]

Mark scheme

(a) **[2]**

- W = m × g = 75 × 10 (1)
- 750 N (1)

(b) **[4]**

Award any four of the following (max 4):

- at the start her speed is zero (low), so air resistance is zero (small) (1)
- the resultant force is downwards (weight is greater than air resistance), so she accelerates (1)
- as her speed increases, the air resistance increases (1)
- so the resultant force decreases and her acceleration decreases (1)
- eventually air resistance equals her weight, so the resultant force is zero (1)
- so there is no acceleration and she falls at a constant (terminal) velocity (1)

(c) **[1]**

- 750 N (1)

ECF from (a)

Worked solution

Weight = 75 kg × 10 N/kg = 750 N. Two forces act on a falling skydiver: her weight downwards and air resistance upwards. When she leaves the helicopter she is barely moving, so air resistance is tiny and the resultant force (almost her whole weight) makes her accelerate. The faster she falls, the bigger the air resistance, so the resultant force and the acceleration get smaller. When air resistance has grown to equal her weight, the forces are balanced, the resultant force is zero and her velocity stays constant: this is terminal velocity. At that point air resistance = weight = 750 N.

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17 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [7]

Mark scheme

(a) **[2]**

- after 20 cm, each 10 cm is fallen in the same time (0.25 s) (1)
- so the ball is moving at a constant speed (velocity), which is the terminal velocity (1)

(b) **[2]**

- speed = distance / time = 0.10 / 0.25 (OR 0.30 / 0.75) (1)
- 0.40 m/s (1)

ALLOW 40 cm/s for the first mark only

(c) **[2]**

- terminal velocity is smaller (1)
- the drag (friction) force from the thicker oil is larger at any given speed, so it balances the weight at a lower speed (1)

(d) **[1]**

- any one from (1):
- film the ball with a video camera (next to a scale) and replay it to read the times
- use light gates connected to a timer
- repeat the experiment and calculate mean times
- use a longer tube with marks further apart

Worked solution

Look at the time taken for each 10 cm: 0.40 s, 0.30 s, 0.25 s, 0.25 s, 0.25 s. The ball speeds up at first, then covers equal distances in equal times, so it has reached a constant (terminal) velocity: 0.10 m ÷ 0.25 s = 0.40 m/s. The ball falls at terminal velocity when the upward drag from the oil balances its weight. A thicker oil gives more drag at the same speed, so the forces balance at a lower speed and the terminal velocity is smaller. Hand timing has a large reaction-time error compared with 0.25 s intervals, so filming or using light gates gives more accurate times.

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18 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [7]

Mark scheme

(a) **[4]**

Award any four of the following (max 4):

- hang the spring from the clamp and measure its original (unstretched) length with the metre rule (1)
- add one slotted mass at a time and measure the new length of the spring each time (1)
- calculate the extension = new length − original length (1)
- calculate each force from the number of masses × 0.50 N (1)
- take readings for at least five different forces (1)
- read the metre rule at eye level OR clamp the metre rule vertically next to the spring OR use a pointer fixed to the bottom of the spring (1)

(b) **[2]**

- independent: the force (weight) applied to the spring (1)
- dependent: the extension of the spring (1)

ALLOW length of the spring for the dependent variable

(c) **[1]**

- any one from (1):
- wear eye protection in case the spring or masses fly off
- keep feet clear of the masses OR put a soft mat (tray) under the masses in case they fall
- clamp the stand to the bench OR use a counterweight so that it does not topple

Worked solution

The force is the variable you change (independent) and the extension is the variable you measure (dependent). Measure the length of the spring with no load, then add 0.50 N masses one at a time, measuring the length each time with the metre rule held vertically beside the spring and read at eye level. Subtract the original length to get each extension, and record force against extension in a table for at least five forces. Stretched springs and heavy masses can fall or spring off, so wear eye protection and keep feet away from under the masses.

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19 · Part 2

Paper 1 & 2 Forces, movement, shape and momentum [7]

Mark scheme

(a) **[1]**

- 6.0 cm (1)

(b) **[3]**

- 4.0 N (1)
- up to 4.0 N, each 1.0 N increase in force increases the length (extension) by 2.0 cm, so extension is proportional to force (1)
- above 4.0 N the increases are 3.0 cm and 4.0 cm, so extension is no longer proportional to force (1)

(c) **[1]**

- 17.0 cm (1)

ALLOW 17 cm

(d) **[2]**

- the spring does not return to its original length (12.0 cm) (1)
- so it has not behaved elastically OR it has been permanently stretched (deformed) (1)

ALLOW it has gone past its elastic limit

Worked solution

Extension = stretched length − original length = 18.0 − 12.0 = 6.0 cm. Hooke's law describes the straight-line (linear) part of a force-extension graph, where extension is proportional to force. Here every extra 1.0 N adds exactly 2.0 cm up to 4.0 N; after that the increases get larger, so the spring stops obeying Hooke's law. At 2.5 N the extension is 2.5 × 2.0 = 5.0 cm, so the length is 12.0 + 5.0 = 17.0 cm. An elastic material recovers its original shape when the force is removed; this spring stays 1.5 cm longer, so it has been permanently deformed.

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20 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [6]

Mark scheme

(a) **[2]**

- no, it does not obey Hooke's law (1)
- extension is not proportional to force, e.g. doubling the force from 2.0 N to 4.0 N more than doubles the extension (3.0 mm to 8.0 mm) OR equal 2.0 N increases in force give increasing extensions (3.0, 5.0, 7.0, 10.0 mm) (1)
- the second mark needs data from the table

(b) **[2]**

- any two from (1) each:
- the original (unstretched) length of the bands
- the thickness of the bands
- the material (type of rubber) of the bands
- the forces (masses) used
- the temperature

(c) **[1]**

- any one from (1):
- read the ruler at eye level (perpendicular to the scale)
- fix a pointer to the bottom of the band and clamp the ruler vertically beside it
- wait until the band stops moving before taking each reading

ALLOW repeat readings and calculate a mean

(d) **[1]**

- a longer wire gives a larger extension, which is easier to measure accurately (1)

ALLOW smaller percentage error OR percentage uncertainty

Worked solution

If the band obeyed Hooke's law, doubling the force would double the extension. From 2.0 N to 4.0 N the extension goes from 3.0 mm to 8.0 mm, more than double, so it does not. In a fair test only the width should change, so keep the original length, thickness, type of rubber and forces the same. Accurate length readings need the eye level with the scale and a steady band. A metal wire stretches very little, so a long wire is used to make the extension large enough to measure with a ruler.

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21 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [6]

Mark scheme

(a) **[1]**

- opposite to the direction of motion of the crate (opposite to the push) (1)

(b) **[3]**

- resultant force = 180 − 130 = 50 N (1)
- a = F / m = 50 / 50 (1)
- 1.0 m/s^2 (1)

ECF for an incorrect resultant force

(c) **[2]**

- the crate keeps moving at a constant speed (velocity) (1)
- because the forces are balanced, so the resultant force is zero and there is no acceleration (1)

REJECT the crate stops OR slows down

Worked solution

Friction always opposes motion, so it acts backwards against the push. The resultant force is the push minus friction: 180 − 130 = 50 N. Acceleration = resultant force ÷ mass = 50 N ÷ 50 kg = 1.0 m/s^2. When the push equals friction, the forces are balanced and the resultant force is zero. Zero resultant force means zero acceleration, not zero velocity: the crate keeps moving at the speed it already had.

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22 · Part 2 Paper 1 & 2 Forces, movement, shape and momentum [6]

Mark scheme

(a) **[4]**

- comb and paper: electrostatic (1)
- apple falling: gravitational (1)
- paper clip and magnet: magnetic (1)
- sliding book: friction (frictional) (1)

ALLOW weight OR gravity for gravitational

(b) **[2]**

- the direction of the car is changing, so its velocity is changing (velocity is a vector) (1)
- a resultant force is needed to change the direction (velocity) of an object (1)

Worked solution

Charged objects attract or repel by electrostatic forces; masses attract by gravitational forces; magnets attract magnetic materials such as steel by magnetic forces; and friction opposes the motion of surfaces sliding over each other. On the circular track the car's speed is constant, but velocity is a vector, so a change in direction is a change in velocity. Forces can change the direction of motion, and a change in velocity needs a resultant force.

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23 · Part 2 Paper 1 & 2 Forces, movement, shape and momentum [6]

Mark scheme

(a) **[1]**

- weight = mass × gravitational field strength OR W = m × g (1)

ALLOW any correct rearrangement

(b) **[2]**

- 9.8 (1)
- N/kg (1)

ALLOW 9.80 OR a value found from a single row of the table

(c) **[2]**

- W = 0.40 × 25 (1)
- 10 N (1)

(d) **[1]**

- to check that the weight is proportional to the mass OR to get a more reliable value of g (by finding the mean of several values) (1)

Worked solution

weight = mass × gravitational field strength, so g = weight ÷ mass. Every row gives the same value, e.g. 4.90 N ÷ 0.50 kg = 9.8 N/kg. The mass of the object is still 0.40 kg on the other planet, but the gravitational field strength is larger, so its weight is 0.40 kg × 25 N/kg = 10 N. Using several masses shows the pattern (weight proportional to mass) and lets you average several values of g, which makes the result more reliable than one reading.

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24 · Part 2

Paper 2 only | Forces, movement, shape and momentum | [7]

Mark scheme

(a) **[1]**

- momentum = mass × velocity OR p = m × v (1)

(b) **[2]**

- p = 0.045 × 60 (1)
- 2.7 kg m/s (1)

(c) **[2]**

- F = 2.7 / 0.0015 (1)
- 1800 N (1)

ECF from (b)

(d) **[2]**

- for the same force, a longer contact time gives a larger change in momentum (force × time) (1)
- the mass is the same, so a larger final momentum means a higher speed (1)

Worked solution

momentum = mass × velocity = 0.045 kg × 60 m/s = 2.7 kg m/s. The ball started at rest, so its change in momentum is also 2.7 kg m/s. Average force = change in momentum ÷ time = 2.7 ÷ 0.0015 = 1800 N. Check: 1800 N × 0.0015 s = 2.7 kg m/s. Rearranging the equation, change in momentum = force × time. Following through keeps the force acting for longer, so the change in momentum is bigger; with the same mass, more momentum means a faster ball.

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25 · Part 2

Paper 2 only | Forces, movement, shape and momentum | [8]

Mark scheme

(a) **[2]**

- change in momentum = 60 × 15 (− 60 × 0) (1)
- 900 kg m/s (1)

IGNORE negative sign

(b) **[2]**

- F = 900 / 0.12 (1)
- 7500 N (1)

ECF from (a)

IGNORE negative sign

(c) **[4]**

Award any four of the following (max 4):

- the seat belt stretches and the airbag squashes (cushions) as the person moves forward (1)
- this increases the time taken for the person to stop (1)
- the change in momentum of the person is the same (1)
- so the rate of change of momentum is smaller (1)
- so the force on the person is smaller (force = change in momentum / time) (1)
- the force is spread over a larger area of the body (1)

Worked solution

The dummy goes from 15 m/s to rest, so its change in momentum is 60 kg × 15 m/s = 900 kg m/s. Average force = change in momentum ÷ time = 900 ÷ 0.12 = 7500 N. Check: 7500 N × 0.12 s = 900 kg m/s. In any crash the person must lose all their momentum; safety features cannot change that. What they can change is the time: a seat belt that stretches slightly and an airbag that squashes make the stopping time longer, so the rate of change of momentum, and so the force on the body, is smaller. Smaller forces cause less injury.

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26 · Part 2

Paper 2 only | Forces, movement, shape and momentum | [7]

Mark scheme

(a) **[2]**

- zero (1)
- both skaters are at rest (velocity zero), so each has zero momentum (1)

(b) **[3]**

- momentum of Q = 40 × 3.0 = 120 kg m/s (to the right) (1)
- total momentum is still zero, so P has 120 kg m/s to the left (1)
- velocity of P = 120 / 60 = 2.0 m/s to the left (1)
- the final mark needs both the value and the direction

(c) **[2]**

- when P pushes Q, Q pushes P with a force (1)
- of equal size and in the opposite direction (1)

Worked solution

Before the push nothing is moving, so the total momentum is zero. Momentum is conserved, so after the push the total must still be zero: Q's momentum to the right (40 kg × 3.0 m/s = 120 kg m/s) is cancelled by P's momentum to the left. P's velocity = 120 kg m/s ÷ 60 kg = 2.0 m/s to the left. The heavier skater moves off more slowly. By Newton's third law, if P pushes on Q, Q pushes back on P with an equal and opposite force, so both skaters feel a force and both move.

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27 · Part 2

Paper 2 only | Forces, movement, shape and momentum | [8]

Mark scheme

(a) **[3]**

- momentum before = 12 000 × 2.5 = 30 000 kg m/s (1)
- total mass after = 12 000 + 8000 = 20 000 kg (1)
- speed = 30 000 / 20 000 = 1.5 m/s (1)

(b) **[3]**

- momentum of Y = 8000 × 1.0 = 8000 kg m/s in the opposite direction to X (1)
- total momentum = 30 000 − 8000 = 22 000 kg m/s (1)
- velocity = 22 000 / 20 000 = 1.1 m/s in the direction truck X was moving (1)
- the final mark needs both the value and the direction

> **ECF** momentum of X from (a)

(c) **[2]**

- truck X exerts a force on truck Y, and truck Y exerts an equal and opposite force on truck X (Newton's third law) (1)
- the forces act for the same time, so the changes in momentum are equal in size and opposite in direction (1)

Worked solution

Total momentum before a collision equals total momentum after, as long as no external force acts. Test 1: 12 000 kg × 2.5 m/s = 30 000 kg m/s, shared by 20 000 kg, so v = 1.5 m/s. Test 2: momentum is a vector, so the momentum of Y counts as negative: 30 000 − 8000 = 22 000 kg m/s, and v = 22 000 ÷ 20 000 = 1.1 m/s, still in X's original direction because X had more momentum. Check: 20 000 × 1.1 = 22 000. During the collision the trucks push on each other with equal and opposite forces for the same time, so whatever momentum X loses, Y gains.

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28 · Part 2

Paper 2 only | Forces, movement, shape and momentum | [7]

Mark scheme

(a) **[2]**

- for an object in equilibrium (balanced) (1)
- the sum of the clockwise moments about the pivot equals the sum of the anticlockwise moments about the pivot (1)

(b) **[3]**

- moment of first child = 300 × 1.6 = 480 Nm (1)
- 400 × d = 480 (1)
- d = 1.2 m (1)

(c) **[2]**

- the weight of the (uniform) plank acts through its centre of gravity, which is at the pivot (1)
- so its distance from the pivot is zero and its moment is zero (1)

Worked solution

For the seesaw to balance, the anticlockwise moment of the first child must equal the clockwise moment of the second. Moment = force × perpendicular distance from the pivot = 300 N × 1.6 m = 480 Nm. So 400 N × d = 480 Nm, giving d = 1.2 m. The heavier child sits closer to the pivot. Check: 400 × 1.2 = 480. A uniform plank pivoted at its centre has its centre of gravity at the pivot, so its weight has no turning effect.

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29 · Part 2

Paper 2 only | Forces, movement, shape and momentum | [6]

Mark scheme

(a) **[3]**

- taking moments about X: 600 × 1.0 = 600 Nm (1)
- F(Y) × 4.0 = 600 (1)
- F(Y) = 150 N (1)

(b) **[1]**

- 600 − 150 = 450 N (1)

ECF from (a)

ALLOW taking moments about Y, 600 × 3.0 = F(X) × 4.0, giving 450 N

(c) **[2]**

- the force from X decreases and the force from Y increases (1)
- the two forces always add up to 600 N OR they are equal (300 N each) when the painter is at the centre (1)

Worked solution

The plank is light, so the only downward force is the painter's weight. Take moments about X, so the unknown force at X has no moment: the painter's clockwise moment is 600 N × 1.0 m = 600 Nm, and the anticlockwise moment of Y is F(Y) × 4.0 m. So F(Y) = 600 ÷ 4.0 = 150 N. The plank is in equilibrium, so the upward forces add up to 600 N, giving F(X) = 450 N. The support nearer the painter takes more of the weight. As he walks towards Y, the force from Y rises and the force from X falls, while their total stays 600 N.

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30 · Part 2

Paper 2 only

Forces, movement, shape and momentum [6]

Mark scheme

(a) **[1]**

- the point through which the weight of an object acts (1)

(b) **[3]**

- moment of the weight = 800 × 0.45 = 360 Nm (1)
- F × 1.2 = 360 (1)
- F = 300 N (1)

(c) **[2]**

- the perpendicular distance of the weight (centre of gravity) from the pivot is smaller, so the moment of the weight is smaller (1)
- so the gardener needs a smaller moment, and at the same distance (1.2 m) this needs a smaller force (1)

Worked solution

All of the weight can be treated as acting at the centre of gravity, 0.45 m from the axle. Its moment about the axle = 800 N × 0.45 m = 360 Nm. To balance, the gardener's moment must also be 360 Nm: F × 1.2 m = 360 Nm, so F = 300 N. Check: 300 × 1.2 = 360. If the load is closer to the wheel, the weight's moment is smaller, so the gardener's force (at the same 1.2 m) can be smaller.

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